

Solve

$$\frac{dy}{dt} = 2y + t$$

[Note: The general form of a first-order, linear ODE is $\frac{dy}{dt} = ay + q(t)$ where $a = \text{constant}$]

Solution with y_p & y_h :

Here: $a(t) = 2$ & $q(t) = t$

Particular solution (y_p):

$$\text{Let } y_p = At + B$$

$$\frac{dy_p}{dt} = 2y_p + t \Rightarrow$$

$$\Rightarrow \frac{d(At+B)}{dt} = 2(At+B) + t$$

$$\Rightarrow \frac{d(At)}{dt} + \frac{dB}{dt} = 2At + t + B$$

$$\Rightarrow A = (2A+1)t + 2B$$

Comparing t & t^0 terms

$$t^0: A = 2B \Rightarrow B = A/2$$

$$t^1: (2A+1) = 0 \Rightarrow A = -\frac{1}{2}$$

$$\Rightarrow y_p = -\frac{t}{2} - \frac{1}{4} \Rightarrow B = -\frac{1}{4}$$

Homogenous solution (y_h):

$$\frac{dy_h}{dt} = 2y_h \Rightarrow \frac{dy_h}{y_h} = 2dt$$

$$\text{Integrating } \int \frac{dy_h}{y_h} = \int 2dt + C$$

$$\Rightarrow \ln(y_h) = 2t + C$$

$$\Rightarrow y_h = Ce^{2t}$$

Complete solution:

$$y(t) = -\frac{1}{4} - \frac{t}{2} + Ce^{2t}$$

Solution using $f(x)$ Rearranging the ODE:

$$\frac{dy}{dt} - 2y = t \quad [\text{Here: } a(t) = -2, q(t) = t]$$

Integrating factor $f(t) = e^{\int a(t) dt}$

$$\int a(t) dt = \int -2 dt = -2t + C$$

$$\therefore e^{\int a(t) dt} = e^{-2t+C} = e^{-2t} e^C = Ke^{-2t}$$

$$\therefore f(t) = Ke^{-2t}$$

Multiplying by $f(x)$ on both sides of ODE:

$$f(t) \frac{dy}{dt} - f(t)y = f(t)q(t)$$

$$\Rightarrow Ke^{-2t} \frac{dy}{dt} - Ke^{-2t}y = Ke^{-2t}t$$

Express LHS as a derivative of a product:

$$\frac{d(e^{-2t}y)}{dt} = e^{-2t}t \Rightarrow d(e^{-2t}y) = e^{-2t}t dt$$

$$\int d(e^{-2t}y) = \int te^{-2t} dt + C$$

$$\Rightarrow e^{-2t}y = -\frac{t}{2}e^{-2t} - \frac{e^{-2t}}{4} + C$$

Dividing both sides by e^{-2t}

$$\Rightarrow \frac{e^{2t}y}{e^{2t}} = -\frac{t}{2} \frac{e^{2t}}{e^{2t}} - \frac{1}{4} \frac{e^{2t}}{e^{2t}} + \frac{C}{e^{2t}}$$

$$\Rightarrow y(t) = -\frac{t}{2} - \frac{1}{4} + Ce^{2t}$$

This term requires integration by parts.

$$\int u dv = uv - \int v du$$

$$\text{Let } u = t \Rightarrow du = dt$$

$$\text{Let } dv = e^{-2t}$$

$$\Rightarrow v = \int dv = \int e^{-2t} = -\frac{e^{-2t}}{2}$$

$$\therefore \int te^{-2t} dt = -\frac{t}{2} e^{-2t} - \int -\frac{e^{-2t}}{2} dt = -\frac{t}{2} e^{-2t} - \frac{e^{-2t}}{4}$$